

FREE MAIL N/C

15-06-2017 Thurs

Ass. FST 224

Engel, Reservation, Description,
Techniques and Applications

~~08/01/473945~~
~~DO~~

hypoxia is a process of preserving
or storing cells, tissues, organs and
other biological materials from any
potential damage by maintaining the
materials at very low temperature
(typically -80°C using solid CO₂ or
liquid N₂ using liquid Nitrogen)

25 June, 2020

Objectives of Cryopreservation

To minimize damage to biological
materials, including tissues,
Materials to be preserved by cryopreservation must

be viable, embryos, tissues and even
entire
as a plan for cryopreservation and
handling of storing, techniques and
applications.

08/01/473945

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very old dry out again
for types 4 dry

figure illustrate the components
of post harvest system for
processing raw materials. A team

1. Cleaning,
2. grading, and
3. Sorting

Many broadly be classified as
This unit of processing operation
consist of series of unit operation.

operation. A processing job
processes is referred to as a unit
by the end users, each of these
final product that is consumed
in converting raw materials to a
product. Many process are involved
characteristics of agricultural
quality or to change the form of
performed to maintain or improve
Define as any activity which is
Post harvest technology may be

POST HARVEST TECHNOLOGY

07/6

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Unit

FINANCIAL

Characteristics of FFCs

MECHANICAL SEPARATION

OPERATION

Most raw materials contains some components which are in-edible or half valuable

Physico characteristics Processing techniques such as sorting, grading, cleaning etc are necessary to obtain the required uniformity of the raw material for further processing.

1. SORTING

Sorting is the separation of raw materials into different physical property. Such as size, shape and color.

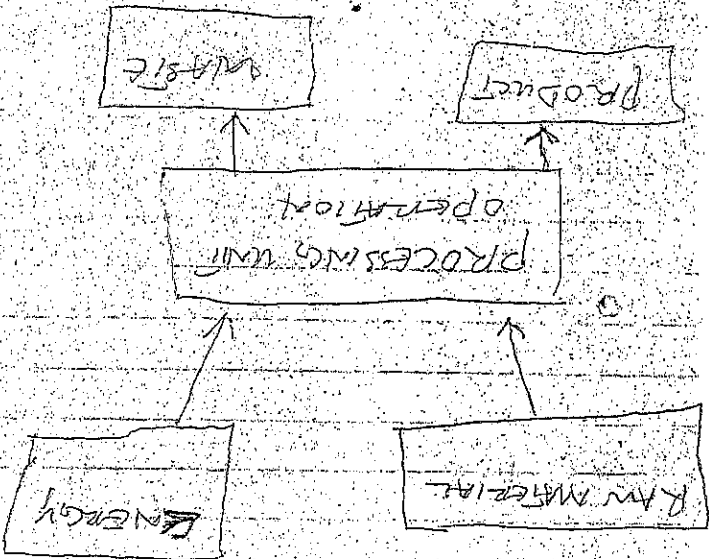
2. CLEANING

Cleaning is the separation operation which involves separating contaminants from the raw materials.

COMPONENTS OF POST HARVEST SYSTEM

Fig. 1

Material is introduced into a processing unit or system. In addition, energy is used and the resulting products is accompanied by waste product.



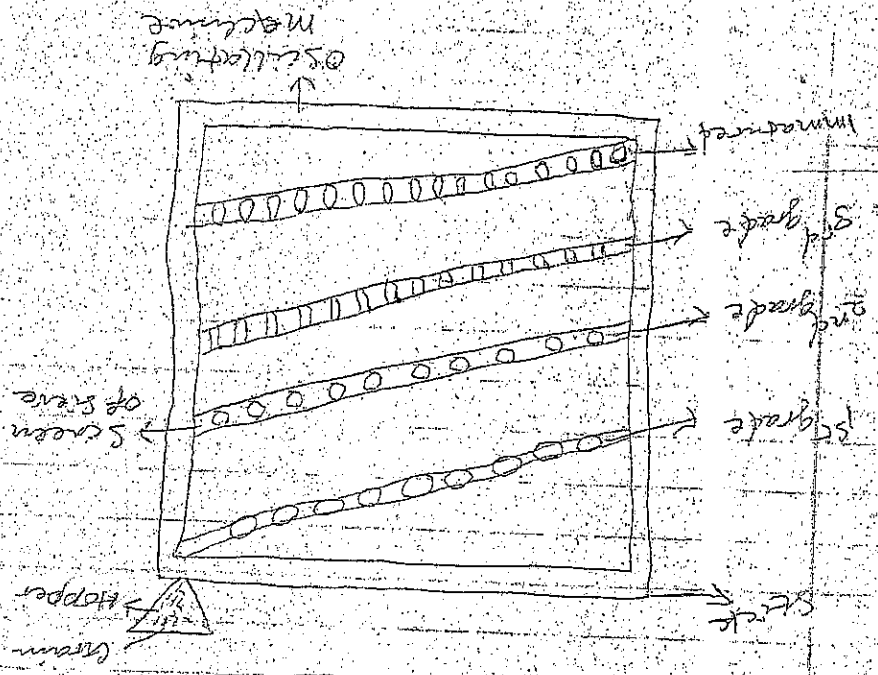
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3/ GRADING

Grading separates raw materials into categories of different qualities. It improves the seed lot by removing of cracks, damaged and shrivelled.

Fig. 2

PRINCIPLES OF OPERATION OF A SCREENING MACHINE



⑤

4/ SORTING MACHINE

Sorting is important because of the following reasons:

1. Sorted produce is better suited to mechanised operation,
2. Sorting is necessary in export situation in which uniformity of heat treatment is required e.g. sterilisation,
3. It gives better control over weight that are filled into standard containers, and
4. Consumers appear to be pleased.

Sorting machine can be grouped into:

1. Weight Sorters,
2. Size Sorters, and
3. Shape Sorters

5/ WEIGHT SORTERS

Weight Sorters are based to utilize the difference in weight

of the product as the basis for sorting. They are gravity machines in which heavier products are separated from the lighter ones.

SIZE SORTERS

They are similar to screening machine. They separate the product according to the size.

SHAPE SORTERS

This grade products according to their shape, thus a sorter for orange can only pick out those product that has the same shape as orange.

GRADING MACHINES

Grading refers to the classification of clean product into various quality depending on the various commercial values and their usage. Grading results into shape separation of materials.

Contaminants

1. The removal of contaminants which constitutes a health hazard, and the control of micro-biological loads and chemical and bio-chemical reaction which impair subsequent process
2. The control of micro-biological loads and chemical and bio-chemical reaction which impair subsequent process

Generally, cleaning of agricultural product have 2 major objectives:

OBJECTIVES OF CLEANING

1. Rice grader
 2. Groundnut grader
 3. Potato grader
 4. Water yam grader
 5. Yam grader etc
- There are many types of graders depending upon the product, (static and cluster at leaves) pressure adjustment of the roller and

independent milling & graded list, grains,

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Effectiveness and produce quality.
Seed Drying

Drying of seed lots i.e. lowering down the seed moisture content to safe

moisture limit is very important

in order to maintain seed viability

and vigor, which may otherwise deteriorate fast due to mold growth

heating and increased moisture -

moisture activity

Importance of seed drying

1. Permit early harvest

2. Permit long term storage

3. Permit more efficient use of land and man power

4. Permit use of plant stock as good fodder

5. Permit seedman to sell a better quality product

15/6/2016

Precaution to be taken during drying

1. Drying of seed to safe moisture

limit should be done earlier

quickly. If the seed are to be

artificially dried they should

be supplied to processing plant soon

after harvesting. Drying is to be done

at the farm the seed should be

evenly spread in the thin layers

either on floor or tarpaulins

3. Care should be taken to ensure

the moisture conditions

does not take place

3. Identity of the lots must be

maintained.

Methods of seed drying

1. the drying & seed flow are the

methods & seeds drying

a. sun drying

b. forced air drying

- Advantages of sun drying
1. It requires no additional apparatus.
 2. It requires no skilled person.
 3. Easy to maintain.
- Disadvantages of sun drying
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- Deep bed drying
- Batch drying

1. Batch - In a batch drying system, the material is dried in a relatively shallow container. Some times

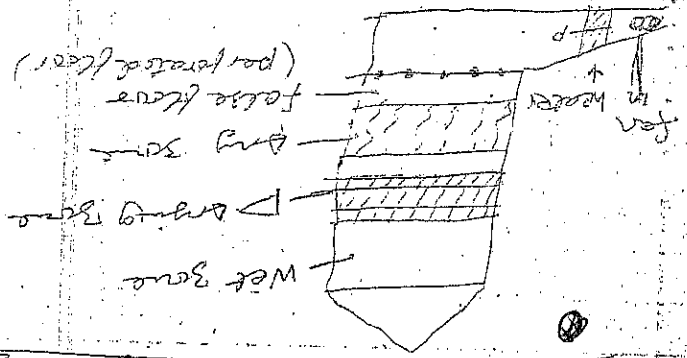
Types of Dryers

Under low pressure, vapour by the application of heat. It is the ^{sublimation} product is fed from by refrigeration. Another method, most time in the freezing.

2. The product, eg water - moisture. Drying is achieved by application of the energy from a radiating heat source.

3. Radiation - Drying is achieved by application of electromagnetic energy usually enclosed in a container. Through the product which is in the container is heated and forced

4. Convection - with stick for turning the products. product, such as driers are equipped



Deep bed dryer showing different zones in figure below.

deep bed drier as a series of shallow The material is stacked into relatively thin layers in which

2. Deep bed dryer -

to another batch.

dry and is removed to give way

is continued until the batch is

Turning an agitator at the drying

There is the allowed to escape

which has collected some moisture

through the product also air

the drying is allowed to pass

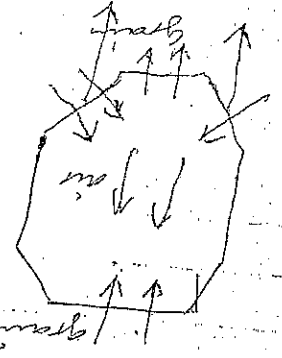
The material is put in trays, and

also sales

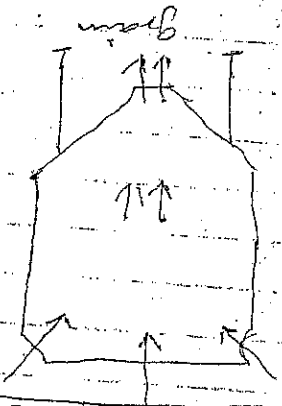
A silo is usually cylindrical in structure at the base and a vertical fan or blower, a loader, plenum and a perforated floor. Hot air is forced through the product and is allowed to pass through the top because of the relatively high depth involved. Drying is not uniform. Rotter there are any point in the time a drying zone existing, drying zone this is caused by moisture migration as air moves through the drying zone.

3. CONTINUOUS FLOOR-DRYER - for more uniform and fast drying, the entire floor dryer is used for both drying air and grain are in motion. The system are used normally.

1. Concurrent and Counter current



CONCURRENT FLOOR DRYER
Both grain and air move in same direction - opposite directions



15. COUNTER CURRENT FLOOR DRYER
In counter current floor drying both the grain and air move in the same direction
concurrent floor drying air and grain

25-6/2016 Separation, separation
is a method of ~~separation~~ processing

agricultural product is used
the material are subjected to
or before they are milled, stored
marketably. The processing are
essential means of getting rid of
foreign materials from the produce.
There are different methods 2

Separation namely:-

1. Separation by specific gravity
2. Separation by dimension
3. Separation by air current
4. Separation by sieves

1. Separation by specific gravity

This method is used locally to get
rid of stone or sand from parboiled
rice. When washing before
cooking, the seed is laid
sprawled in a stream of water,
stones and other heavy objects

(17)

Sink to the bottom.
2. Separation by direction: This
may be due to difference in weight,
length, slope, width. The equipment
used are bar sieve separator and
sieves.

3. Separation by air current: This
take advantage of differences
of terminal velocity of materials.
Asperator is an equipment
used for separation by air
current. Separation by ascending
air current to separate dust,
chaffs, small straw particles and
lighter seed from sand grain.

4. Separation by sieves: Separator
can be carried out with different types
of sieve known as separator
sieve. The different types of
separator sieve are classified
as follows:

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a. Flat Seives

b. Rotary Seives

c. Oscillating Seives.

DISINTEGRATING MACHINE

Disintegrating is a breaking up of particles into smaller particles.

As a result of impacts when a

material is subjected to sudden

blow or force in absence of stress

strength. Cracking of rocks with the

help of hammer. Operation of

hammer mill is an example of

dynamic force application by

impact methods. There are

many disintegrating methods

equipments but only two have

be discussed in this course.

The two disintegrating

equipments are:

1. Hammer mill

2. Rotary mill or disintegrator

The Hammer mill consists of

rotary shaft with a rotor

(19)

Hammer mill

The hammer mill consists of

essentially of a rotor carrying

a rotor that bears a number

of hammers and its position

periphery in the diagram. The

weighting is in circular but it

can also be rectangular. A

breaker plate is used to enhance

comminution, when a screen

is used to receive the material

bigger than the nominal size

of the product. The size of the

product can be changed by

rotor speed, feed rate, clear

ance between hammers and

grading plate numbers and

size of the hammers and the

size of the screen. The disintegrating

range of the hammer mill including

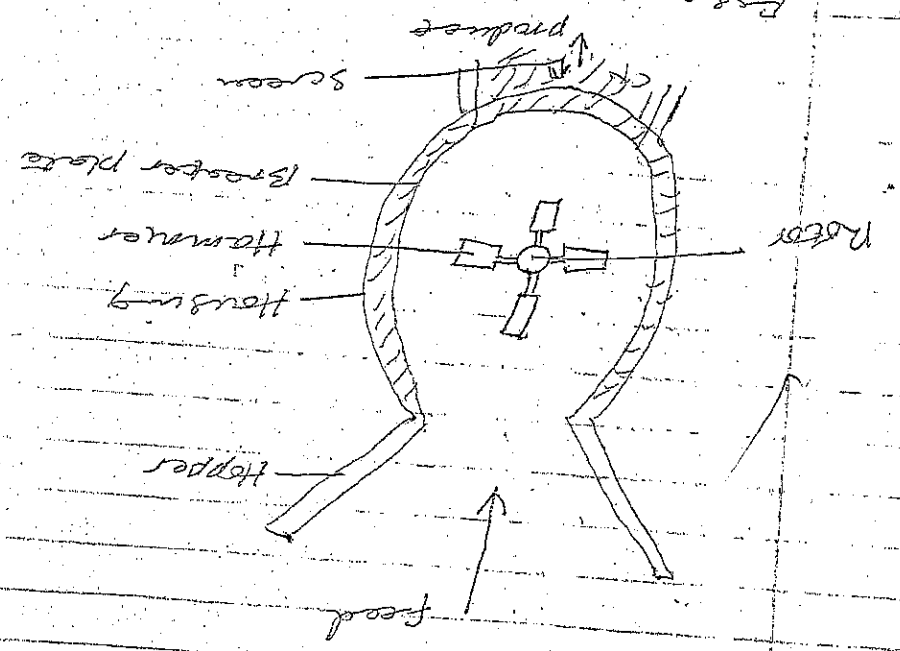
large power requirements and

and some vibration grading.

(1) In change between power requirement

However the advantages are -
 And freedom from damage
 by foreign material for
~~essential~~ The operation of
 of hammer mill can be seen
 below.

Essential features of a Mill Hammer



R.H.P.2. mill or disc mill
 P.B.T.s will consist of rotor
 inside a circular screen.
 The rotating shaft is usually
 vertical. The rotor includes
 a number of hammers rotating
 round at a fairly close
 clearance. The hammers are
 generally fixed to the shaft.
 Part in some cases giving
 hammer are also used. Feed
 enters the mill chamber
 parallel to axis in the disc
 type attrition mill. The
 product is discharged out
 through a perforated bagging
 screen which serves as the
 rotor. ~~the diagram~~ of P.B.T.s mills -
 many applications are in
 used materials. The ~~are~~ collecting
 e.g. this mill uses bond in

These cases where solid content
 is in the range of 40-50% more
 Rotary mill is able to grind material
 below 15 micron size.
 Rotary mill are made in
 normally supplied in
 rotors diameters from 10-60 cm
 diameter (4-24 in dia). The
 number tips spread in the range
 of 5-2-11m per second and have
 power ranges from 0.5-200 hp

(22)

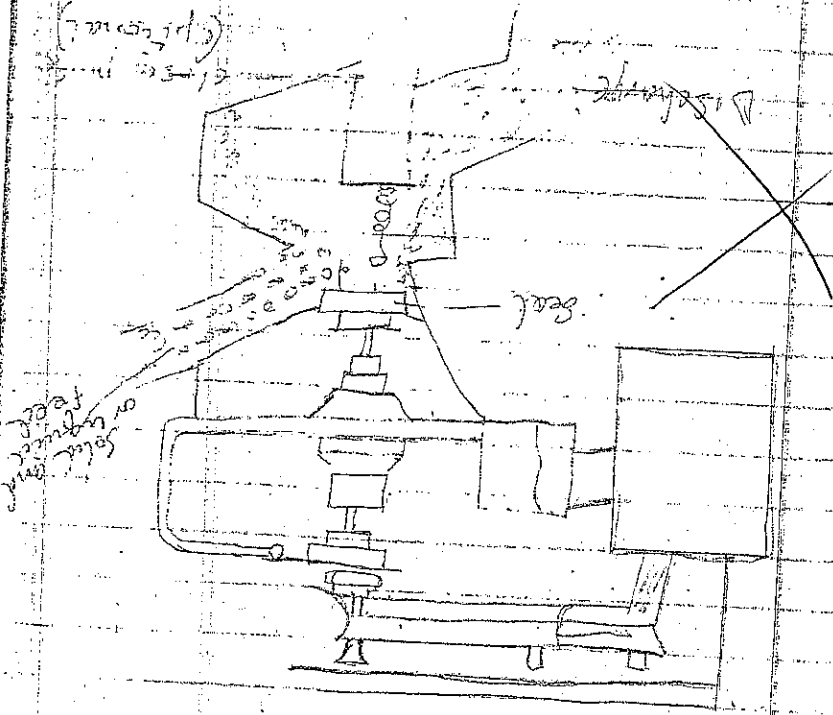
X

X

Essential features of a rotary mill

(23)

X



(24)

Discharge

Seal

Solid feed
or liquid feed

Discharge

STORAGE

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Definition of storage: Storage is an intermediate and repeated phase during transit of agricultural products from producer to processor and its products from processor to consumer. Beside agricultural product there is need to be stored from one harvest to the next thus, demanding additional carry over as to safeguard against lowering crop of low yield or poor quality or against speculation in price and market demand.

In Nigeria, storage is even more important because the storage system in place are not fully developed. By some estimate up to 30% food produce in this country is lost due to poor storage condition. This is worse of fruits storage and vegetables.

Me

END

IMPORTANCE OF STORAGE

(25)

1. To keep planting materials from one generation to another without loss of viability.
2. To ensure steady supply of food throughout the year, since food harvest are seasonal.
3. To stabilize food price by stock piling when food price are low and releasing them into the market when the price go up.
4. To encourage farmers to produce more food since the excess of what they produce will not go on loss.

✓ CONTROLLED CONDITIONS OF STORAGE

the factors to be controlled -

Storage are as follows:

1. Temperature
2. Relative Humidity
3. Time of Storage or Shelf Life
4. Moisture Content

Temperature is one of the most important factors affecting the storage of agricultural products quality. Higher temperature accelerate flower development and senescences, lower temperature slow down the respiration rate and the utilization of carbohydrates and other storage materials in plant tissues.

2. RELATIVE HUMIDITY

Relative humidity influence the survival of insect mainly through

(24)

Survival

Through the effect on their water content. Dry conditions are generally unfavorable for the reproduction of most insects. Time of storage on shelf life. Shelf life or storage life is a length of time for which a item remains able to be eaten, used for seed, for increasing seed storage period for some fruit and vegetable, reducing the oxygen level in the storage air and increasing the carbon dioxide level as a supplement to respiration can provide extended storage life for this commodity under the controlling conditions of storage requirements and preservation of fruit and

(25)

Proton Pocket (29)

Factors affecting storage life of vegetables and fruits

The factors are the factors that affect the storage life of veg-
etables and fruits:

1. Temperature
2. Water loss
3. Mechanical damage
4. Decay - storage

1. Temperature

Temperature is one of the factors affecting storage life of vegetables and fruits, higher temperature accelerates

growth development and senescence. But when the

storage temperature of about 4°C and below, the

85-90% are maintained, the

vegetables and fruits remain in

most desirable condition

for use.

Storage requirements and the properties of fruit and vegetable

vegetables are indicated in the table below:

Commodity	Type	Temp °C	RH %	Approximate shelf life weeks
1. Frozen packed fruit	Fruit	(-23) - (-17)	-	26-53 wks
2. Frozen packed vegetables	Veg. products	(-23) - (-17)	-	26-52 wks
3. Fresh fruits	Fruit	13-14	85-90%	4-19 wks
4. Beans	Beans	0-4	85-90%	2-4 wks
5. Sweet potatoes	Sweet potatoes	13-16	95-80%	17-36 wks
6. Lettuce	Lettuce	0	90-95%	2-3 wks

you can break

to

fill

the

table

In short low temperature prolongs the storage of fruit and vegetables. 2. Water loss; fruit and vegetables are generally covered or sealed by high moisture content, say between 85-90% water bases. As a result of the higher moisture content of the product the shelf life is short and longer special environment is required to improve the shelf life. Fresh fruits and vegetables when it test the moisture content, the shelf life is reduced or decreased, however, the loss of only the few percent of the moisture content, the toughness of the product begins to rather rather and its resistance to shelf damage and shelf life decrease.

(30)

3. Mechanical damage; sometimes and containers may also cause damage to fruits and vegetables by causing tunnelling in with pores resulting in the rotting of the structure of the containers which may be affect the product inside. Mechanical damage is also due to impact during transport or creation product from transportation facilities. 4. Decay in storage; must from product begin to under go deterioration as seen as they are known (rot) the agent of spoilage or decay - check

(31)

33
This is a maggot / 1st / 2nd / 3rd / 4th / 5th / 6th / 7th / 8th / 9th / 10th / 11th / 12th / 13th / 14th / 15th / 16th / 17th / 18th / 19th / 20th / 21st / 22nd / 23rd / 24th / 25th / 26th / 27th / 28th / 29th / 30th / 31st / 32nd / 33rd / 34th / 35th / 36th / 37th / 38th / 39th / 40th / 41st / 42nd / 43rd / 44th / 45th / 46th / 47th / 48th / 49th / 50th / 51st / 52nd / 53rd / 54th / 55th / 56th / 57th / 58th / 59th / 60th / 61st / 62nd / 63rd / 64th / 65th / 66th / 67th / 68th / 69th / 70th / 71st / 72nd / 73rd / 74th / 75th / 76th / 77th / 78th / 79th / 80th / 81st / 82nd / 83rd / 84th / 85th / 86th / 87th / 88th / 89th / 90th / 91st / 92nd / 93rd / 94th / 95th / 96th / 97th / 98th / 99th / 100th

engines, environmental conditions, insects and rodents, micro-organisms and metabolism.
Decay in storage reduction.
Short life of storage fruits
and vegetables for long quality
times storage of fruits and
vegetables for basic require-
ments must be made. Most
winter maintenance of
optimal temperature
2. Preservation of winter
content

POST HARVEST FACTORS THAT
INFLUENCE DECAY IN
STORAGE

1. Engines
2. Micro-organisms

There are two major causes
of storage decay in storage

- 33
1. Organs of storage
 2. Edible reproductive parts
 3. Frost/fruits and vegetables
- Basically explain the following
in relation to storage

Engines: many food products
substance which make the
decay or spoil with age.
This substance are changed
engines contacts all the basic
processes of living things. In fruits,
engines causes them to ripen
and this process goes on
leading to over ripening and
breaking down which results in
decay.

2. Micro-organisms: the second
major causes of rot in fruits

Post

is attack by micro-organisms
there are two main types of
micro-organisms responsible
for decaying of food spoilage.
These are:-

1. Bacteria

2. Fungi

3. Bacteria

Bacteria is often referred to as
germs, bacteria grow quickly
in warm and damp conditions.
Some are very harmful and
cause food poisoning if eaten
by man.

2. Fungi, like in warm and
damp conditions fungi produce
microscopic food called
spores which are carried in
air and water. When these
lands, it grows quickly. There
are two types of fungi i.e. mold

(35)

and yeast. It yeast get into

warm place, it can cause

fruit fruit and drink to

ferment.

→ MOLD: While mold get into

contact with food it makes

the food to be coloring and

black.

STORAGE STRUCTURES

~~Bacteria~~; Uncertainty of

weather conditions prevailing

on the surface of earth

affect the total production of

grains and other food material

- is. Here, all countries will

have to maintain a buffer

stock of food grains, fruits

and vegetables, feed and

fodder, sugar and cooking oil etc

to sustain a human and animal

production.

1. It should maintain the permanent
we design to meet the future
program ~~it~~ ^{it} ~~just~~ well, it must
 2. It should provide protection
from insect, birds, mice
and rodents.
 3. It should have easy
operation facilities
 4. It should have facilities
for accurate and fast gate
 5. It should be economical
 6. It should permit cleaning
- These materials are to be stored
in small godowns, warehouses,
cups, silos, coal storage and
open yard etc.
- REQUIREMENT OF A GOOD
STORAGE STRUCTURES
- For a storage structure to

7. It should protect the product
from fire and theft.
 8. It should have good loading
and off loading facilities.
- VENTILATED STORAGE
STRUCTURES
1. Cribbing are highly
ventilated structures for
storage of guns, machine guns,
ammunition, etc. and they
are stored at low humidity.
If the humidity is high, the
weight of such storage struc-
tures will be highly limited.
The width can be up to 6m
in the northern part of the
country ^{where} ~~where~~ humidity
is less and 1m in the
southern part where
humidity is high. The
additional types of crib

has lower level of ventilation
and therefore requires

product to remain much

longer in the feed before

removing it. This is a most

important matter, the correct

cost may be low

depending on the method

and locality.

2. Team bands; in fact not band

around, you are storing bands.

The young may be cured before

being put in the band, the

important features of these

bands are:

1. Good ventilation - air and

the young.

2. Speed for the heat of the sun

3. Security against poisons

and

Only good young with no rot

should be kept in the band,

as well as should be regularly

inspected for spotted young

and for pests, especially rats.

A strong angle is made by

planting post, as well as a fast

growing trees; the post sprout

branches and leaves used

provide shade. If rats are a

problem, they can be kept out

by constructing a corrugated

iron sheet around the bottom

of the upright poles corner

climb up corrugated sheet

then young are raised up to

make it more difficult for

rats, tegants and other pests

to reach them.

3. Cold storage; this is a method

of preserving perishable

conditions in their feed form and whole some for a longer period by controlling temperature and humidity condition within the storage compartments. Cold storage comprises the effect of low temperature in retarding or arresting the activities of bacteria that cause spoilage of milk. Also cold storage is the method of storing fruits and vegetables as in cold and humid conditions that are constantly maintained by refrigerator and by liquefying hydrocolloids may be regarded as a second dry refrigeration in which ice crystallizes after freezing is used to cool a water based contents. The main objects of cold storage are:



1. To maintain high productivity and thereby check deterioration and maintain fertility.

2. To maintain low temperature (23°-16°C) and sex ratio monitor problems and maintain highest check activities of micro-organisms agents of degeneration. The problems of fertility of fertility is now be broken down in 1st at 32°C after breeding at 10°C and 1 month 0°C.

4. ~~Referring Referring Referring~~ AND

5. ~~Referring Referring Referring~~

1772

(47)

ice which keeps food fresh for 2-3 days. freezer should have a temperature at least -18°C and some go as low as -25°C . most fruit and vegetables have to be washed before freezing. Take bacon & sausages. Large food can be kept -18°C .

MONTHS

- 1. Best 12-15 months
- 2. Fruit & vegetables 10-12 "
- 3. Eggs, poultry, meat 8-10 "
- 4. Cheeses, butter fresh pork 6-8 months
- 5. Fish 4-6 months
- 6. Bacon, uncooked 2-4 months

Food

PRODUCTS

(48)

The melting of frozen food before cooking is called thawing & defrosting.

Food preservation:

Food preservation can be done in a number of ways, but the most common is drying or storage of food, thus allowing it to be stored in a but suitable for human use. Importance of food preservation:

1-

Increasing the shelf life of food, thus increasing the supply

2. Making the seasonal food available throughout the year

3. Adding variety to the diet

4. Saving time by reducing preparation

At the end of essay

④

Food Spoilage

Food spoils, due to deterioration changes that occur in it, that makes it unfit for use or unpalatable. Food changes from the original state or structure. This change may result in making the food unfit for use or being poisonous. Several causes for food spoilage are:

1. CAUSES FOR FOOD SPOILAGE:
 growth of microorganisms which bring about undesirable changes.
 Action of enzymes present in the food.
 3. Oxidative reaction in the food.
 4. Mechanical damage to the food (e.g. bruising of bananas; mashing, etc.).
 5. Damage due to port (e.g. insect and rodents).

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Microbiology in Food

METHODS OF FOOD PRESERVATION

Food preservation is a method of preventing or delaying the growth of microorganisms which microorganisms are unable to grow in the food.

1. Bacteriostatic
 2. Bactericidal
 3. Bacteriostatic
 4. Bactericidal

Bacteriostatic method is which microorganisms are unable to grow in the food. e.g. in drying, pickling, salt, oil and spices, chemical preservatives etc.

Bactericidal method is which microorganisms are able to grow in the food. e.g. pickling, salt, oil and spices, chemical preservatives etc.

2. Bactericidal
 3. Bacteriostatic
 4. Bactericidal

DEHYDRATION
 Drying of food is a method of preventing or delaying the growth of microorganisms which microorganisms are unable to grow in the food. e.g. in drying, pickling, salt, oil and spices, chemical preservatives etc.

in the food ~~and~~ is removed
and the concentration of water
in the food is lowered by
the addition of a certain amount of
the insoluble to ground and
the food. Dehydration is the
removal of H₂O from the product
and this can be done by:

1. Drying: in a tray or container
it and this can be done by:

Direct drying of the food
for drying a variety of food,
vegetables and fruits, this can be
used, dried, prepared and
placed on flat bottom tray
under the sun.

2. Smoking: food can also be
dried by exposing it to
smoke by burning some special
kinds of woods, in this method
the food from the smoke helps
in removal of the moisture
exposure to smoke & heat

a characteristic flavor to the
food. Fish and meat are the
food usually preserved by this
method

3. Addition of salt or sugar, in
up (boiling) moisture by addition
of salt such as salt and
sugar will supplement growth
of micro-organisms. Growth and
moulds are prepared by
boiling food fruit pulp or fruit
peels with a sufficient quantity
of sugar (about 55% of sugar)
to a sufficiently thick consistency
and help enough to hold
the fruit tissue in position.

The high concentration of
sugar and other solids binds
moisture and making it unavailable
for micro-organisms to grow

(48)

4. Use of oil: A layer of oil on top of any fruit & veg

Prevent the growth of micro-organisms on or fruit by preventing exposure

Use of chemical preservatives

Controlled climates when added

in small quantities can hinder undesirable chemical reactions

is interfering with the cell membrane in food by

use of the micro-organisms, with enzymes acting on their genetic mechanism.

ii) Acting as anti-oxidants:

Maximum amount allowed to be added to each type of food

is regulated by law because high concentration can be

harmful. Some of the chemicals used

Chemicals preservatives used

(49)

1. Sodium benzoate
2. Calcium propionate
3. Sorbic acid

Insect Pest

pest in any organism which

action present near, in or on

crop plants, has a direct or

indirect effect on the

growth, productivity or produce of the crop plants

Wangt controlling insect pest in the store.

1. Sanitation
2. Proofing
3. Mechanical control
4. Chemical control
5. Cultural method
6. Biological method

Basically discuss why insects pest can be determined (steps)?

pest and disease cause a substantial loss to crop farms. Other than reducing yield quality, pest and disease control increases the production cost in terms of buying the chemical to control them. It has been estimated that about 40% of world production is lost due to action of pest and diseases. After harvest, an additional 25% may be lost during storage and transportation.

Aside from the economic loss, they can cause, pests are ~~well known~~ vectors for disease, may contaminate produce and cause structural damage to facilities.